

Deutsche Bank
Research

ASSET TOKENIZATION 101

The Next Phase of Financial Infrastructure

September 2026

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Deutsche Bank Research Institute

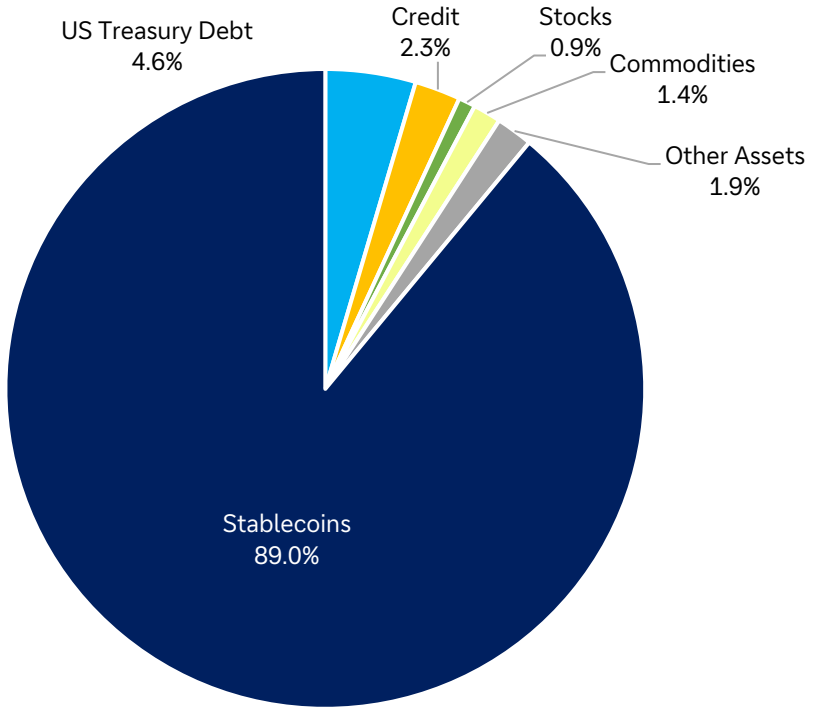
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EXECUTIVE SUMMARY: Tokenization expands into market infrastructure

- The global tokenized assets market, excluding stablecoins, grew from ~$10bn in Jan 2025 to ~$39bn in Sep 2026. US Treasury debt remains the largest segment outside stablecoins, with money market funds leading near-term adoption. Despite the US Digital Asset Market Clarity Act failing to advance through to the Senate yesterday, we remain positive about the momentum to tokenize both assets and capital markets in the coming years.
- Institutions are increasingly applying tokenization to settlement, funding and collateral management. Notable examples include DTCC's tokenization service, DLT-based intraday repo and collateral initiatives, and the Eurosystem's Pontes solution, which connects DLT platforms to TARGET Services and is expected to launch later this month.
- A broad tokenized capital market requires progress across three areas: assets, settlement money and market infrastructure. The tokenized RWA market remains small relative to traditional markets, and token issuance does not automatically create secondary-market liquidity.
- In this chartbook, we broaden our focus from tokenized assets, such as bonds and stocks, to the infrastructure through which they are issued; and review (i) market size; (ii) adoption trends; (iii) the regulatory drivers of tokenization; and (iv) barriers that will determine the pace of growth.

Market share of tokenized assets, Sep 14 2026



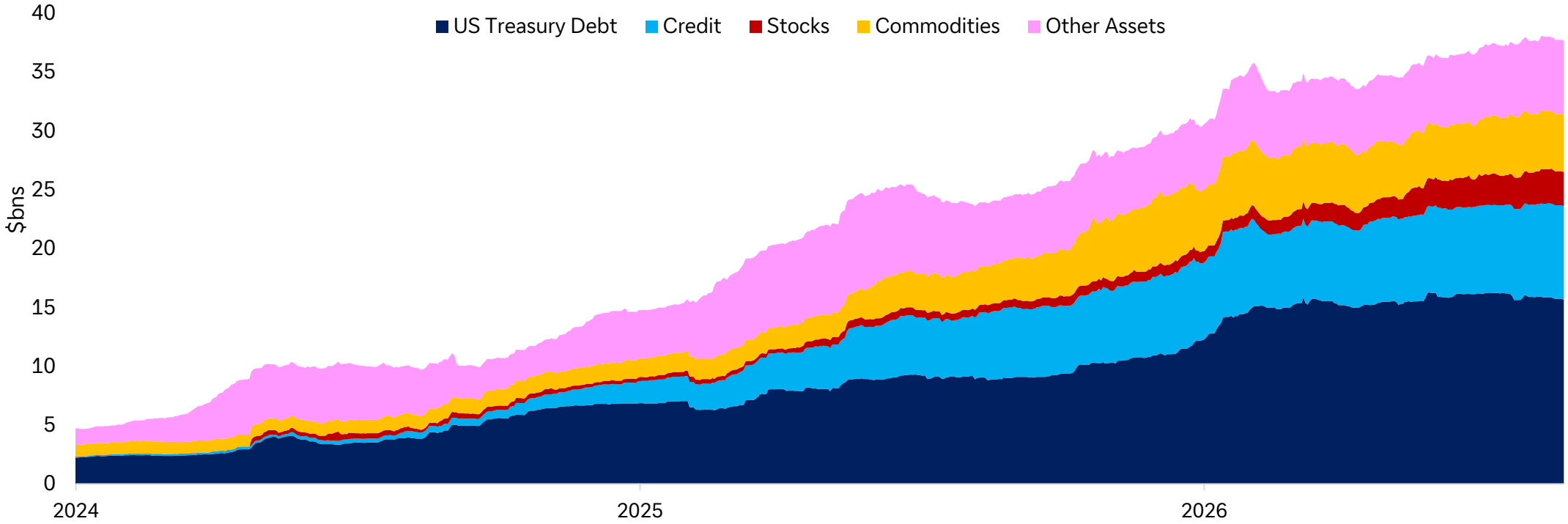
Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" denotes tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Note: Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, private equity, venture capital and real estate.

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MARKET SIZE: US Treasuries continue to dominate

- US Treasury debt remains the largest on-chain tokenized asset segment excluding stablecoins, reaching $15bn in September 2026. Much of that is driven by tokenized money market funds (MMF).
- Demand reflects appetite for highly liquid, short-term risk-free collateral assets, with the ability to generate yield.

Distributed tokenized asset value, excluding stablecoins



Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" refers to tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, stocks, private equity, venture capital and real estate.

DEFINITION : How asset tokenization works

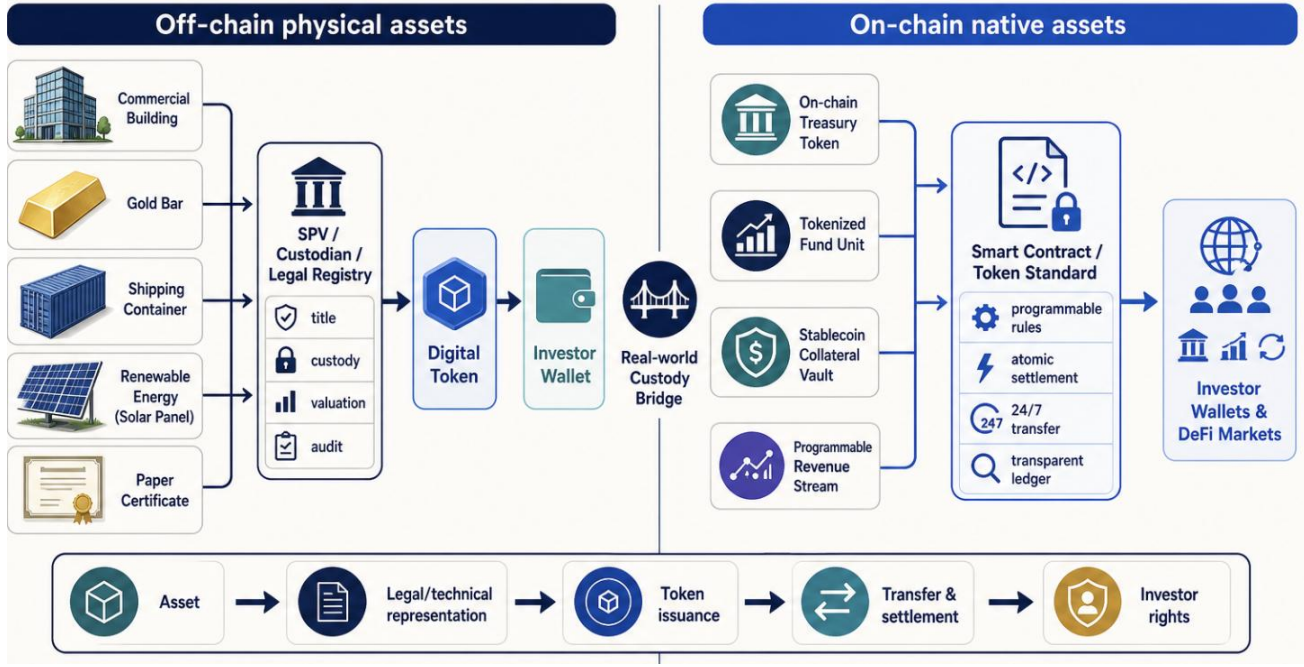
- Tokenization records assets or claims as digital tokens on a distributed ledger, using public or permissioned networks.
- Potential benefits include faster transfers, automated servicing and smaller investment units.
- Shared records can reduce reconciliation, but trading hours, redemption terms and investor access remain product-specific.

Two issuance models:

- On-chain (or native): The security or fund share is issued directly on a distributed ledger.
- Off-chain (or tokenized representation): The underlying asset remains outside the blockchain, but ownership claims and economic rights are represented digitally on-chain.

Component	Traditional form	Tokenized arrangement
Settlement assets	Bank deposits, central bank reserves, cash	Stablecoins, tokenized deposits, CBDCs, central bank money available through DLT platforms
Issuance assets	Treasury bills, government bonds, money market funds, real estate, commodities etc	Tokens representing securities, fund shares or claims on underlying assets
Market infrastructure	Custodians, exchanges, clearing houses and transfer agents	DLT-based issuance, settlement and servicing, often operated by existing institutions

Tokenization can enable automated transfers and settlement



Native digital securities are issued on-chain but may still require custodians, registrars and other intermediaries, according to the SEC.

Sources: Deutsche Bank Research. Image generated by AI.

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STABLECOIN SPOTLIGHT: The settlement side of tokenization

- Stablecoins account for ~89% of the combined value of stablecoin and distributed tokenized assets. We assess them separately because their principal role is payment and settlement.
- They support near-instant, round-the-clock transfers for remittances, business payments and treasury operations. Cross-border savings depend on transaction fees, FX costs, compliance and conversion into bank money.
- Chainalysis projects adjusted annual transfer volume could rise from $28tn in 2025 to $719tn by 2035; higher scenario approaching $1.5 quadrillion.

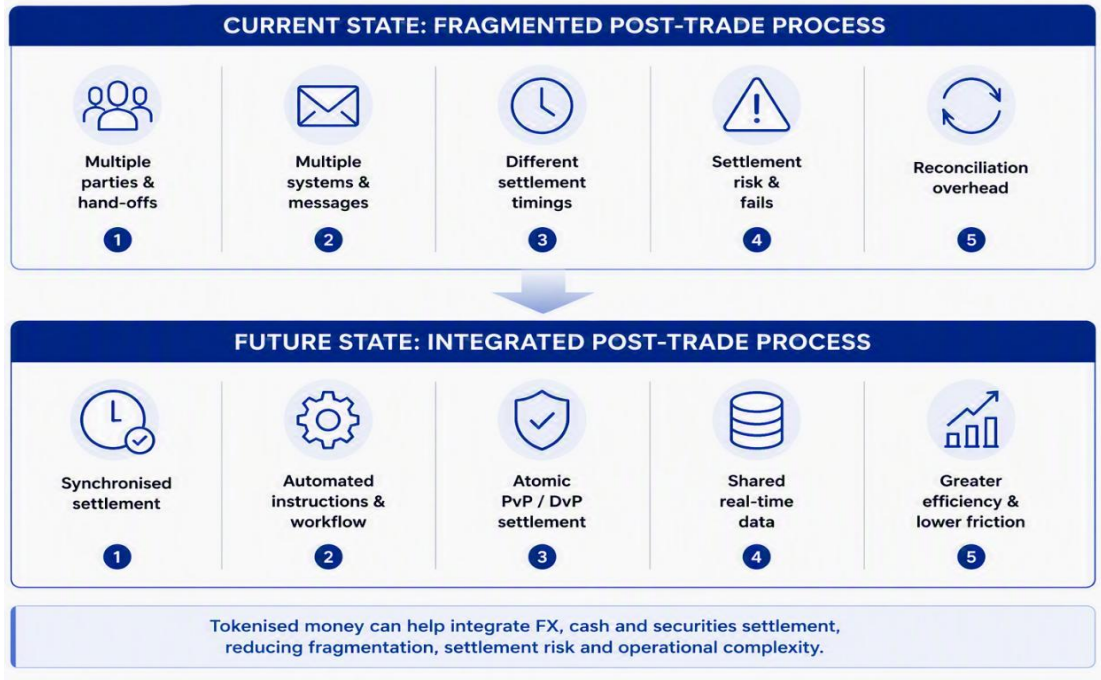
Regulation and competition

- The US Treasury has proposed implementing rules for payment stablecoins. The GENIUS Act takes effect on the earlier of 18 January, 2027, or 120 days after final implementing regulations are issued.
- Stablecoins could increase competition for bank deposits and payments. Exchange rewards are distinct from issuer-paid interest, which GENIUS prohibits for permitted payment stablecoins.

Risks

- Trust depends on reserve quality, legal redemption rights and operational resilience. Stablecoins remain dependent on their issuers and supporting institutions.

A more integrated post-trade process



Sources: Deutsche Bank Research, taken from DB's FX Special Report: Tokenised money and the future of FX - Deutsche Bank Research

Tokenized deposits and central bank money provide alternative settlement options. Connecting the cash and securities legs of a transaction can improve settlement efficiency and reduce fragmentation.
For more on tokenized money and settlement, particularly for FX, read here.

INSTITUTIONAL ADOPTION: Tokenization expands into funding, settlement and collateral

- Early efforts focused on tokenized claims on traditional securities (i.e. BlackRock's BUIDL, Franklin Templeton FOBXX), while the underlying assets remained off-chain.
- Financial institutions are now extending tokenization to the infrastructure through which assets are distributed, traded, settled and financed.

DLT adoption: small, but intraday repo offers a practical use case

- Global DLT fixed-income issuance reached €4.8bn in 2025, up 48% from 2024.
- DLT intraday repo (i.e. JPM Kinexys and Broadridge's DLR) can accelerate collateral transfers and settlement, helping institutions manage short-term funding needs.
- A Finadium study estimated that using intraday DLR for 15% of activity could reduce intraday liquidity buffer needs by 8–17%.

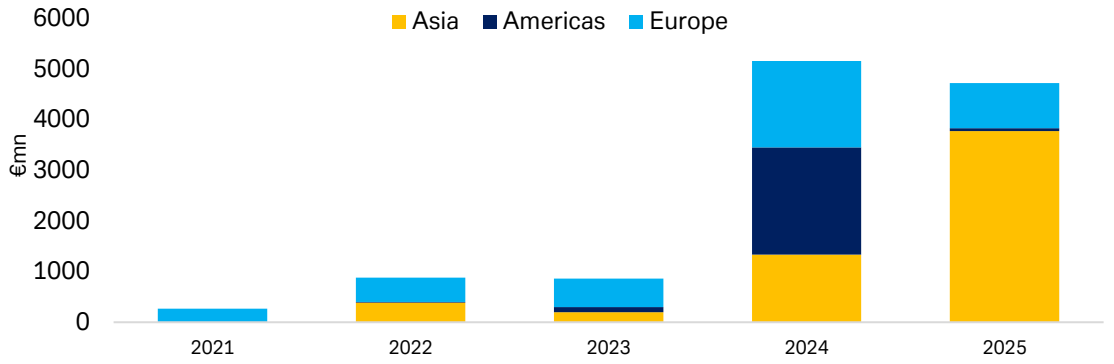
Tokenized securities on blockchain rails

- DTCC to launch its tokenization service in Oct 2026, covering Russell 1000 constituents, ETFs tracking major indices, and US Treasury bills, notes and bonds.
- Initial production trades took place in July with >30 participating firms. Tokens are recorded on DTCC's Hyperledger Besu network or on Canton, settling on either. The SEC granted a three-year authorization.

Official-sector initiatives and interoperability

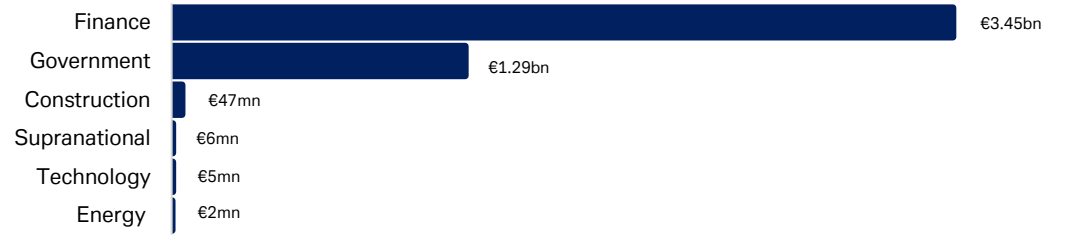
- Project Agorá, led by the BIS and IIF with central banks and financial institutions, demonstrated a prototype for cross-border settlement using tokenized deposits and central bank reserves. Real-value testing is the next phase.
- Clearstream, DTCC and Euroclear published a joint interoperability framework with BCG in March 2026 to address fragmentation across digital asset markets.
- Eurosystem's Pontes solution connects DLT platforms to TARGET Services for settlement in central bank money.

Digital bond issuance by region



Source: Deutsche Bank Research, Association of Financial Markets Europe.

Global digital bond issuance by issuer type in 2025



Source: Deutsche Bank Research, Association of Financial Markets Europe.

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SHORT-TERM ADOPTION: Tokenized Treasuries and money market funds lead

- Tokenized cash-management funds, including BUIDL and USYC, combine short-term yields with more flexible transfers and collateral use.
- Potential benefits include faster settlement and lower transaction costs, making Treasury exposure more accessible across eligible digital platforms.
- Direct Treasury bond tokenization accounted for 27% of overall US Treasury tokenization in 2025, indicating adoption beyond fund structures.
- These assets can also serve as collateral in repo and margin arrangement, supporting demand beyond investment returns.

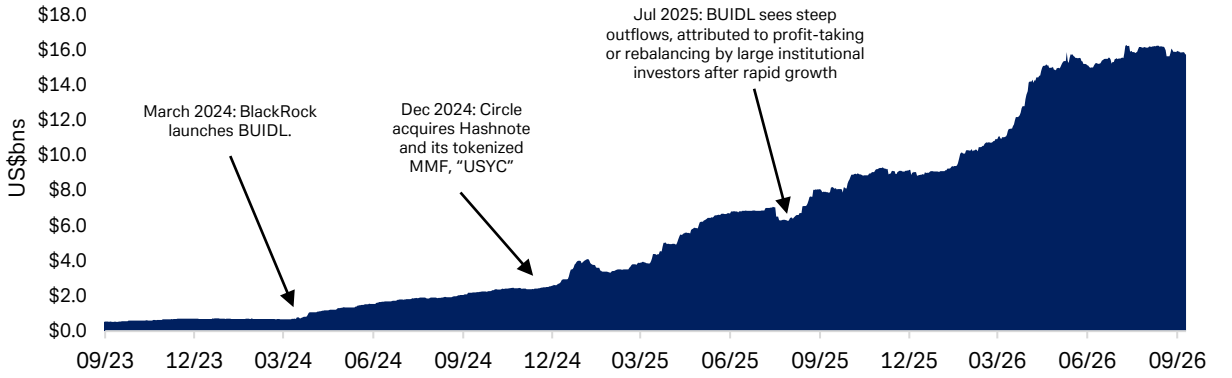
Secondary-market constraints

- Tokenized bonds are often held to maturity. Limited investor participation and weak interoperability between platforms constrain secondary-market liquidity.
- Some exchanges are attempting to improve liquidity by connecting digital and conventional markets through dual listings, including SIX digital exchange and Swiss Stock exchange).

Risks

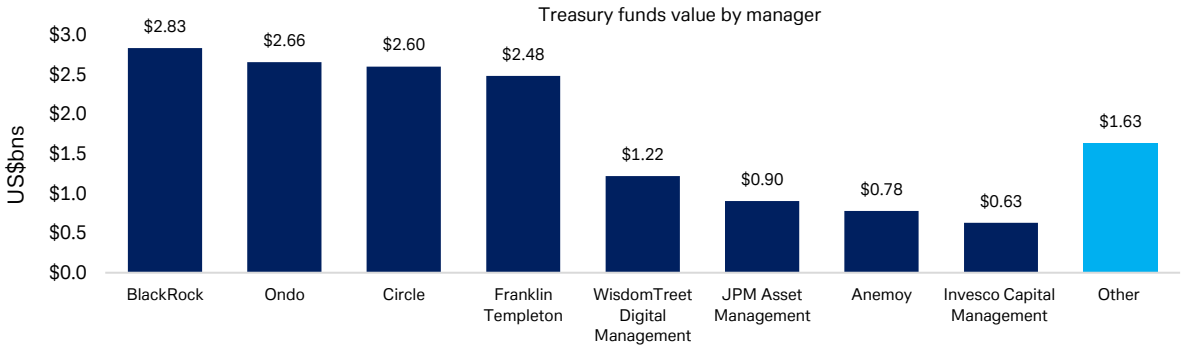
- Redemption terms can differ from the settlement hours and liquidity of underlying assets, creating potential liquidity mismatches.
- Blockchain transparency can improve visibility but may also accelerate withdrawals during stress.
- Using tokenized fund shares across multiple products and as collateral can create additional channels of contagion.

Tokenized US Treasury market value reaches $15bn



Source: Deutsche Bank Research, RWA Database. Updated Sep 10, 2026.

Largest providers of tokenized Treasury products



Source: Deutsche Bank Research, RWA.xyz. Data from September 10, 2026. TMMF asset class includes tokenized US treasuries, corporate bonds, and non-US government debt.

MEDIUM-TERM ADOPTION: Tokenized equities remain small; DTCC's launch could broaden adoption

- Tokenized equities remain small (<1%) relative to traditional stock markets.
- 68% of equity tokenization in 2025 consisted of direct tokenization of individual stocks or baskets of stocks, vs. tokenized equity funds.
- Suggests that blockchain adoption is increasingly focused on the securities themselves (i.e. DTCC method) rather than through third-party platforms (i.e. Ondo and xStocks).

Developments

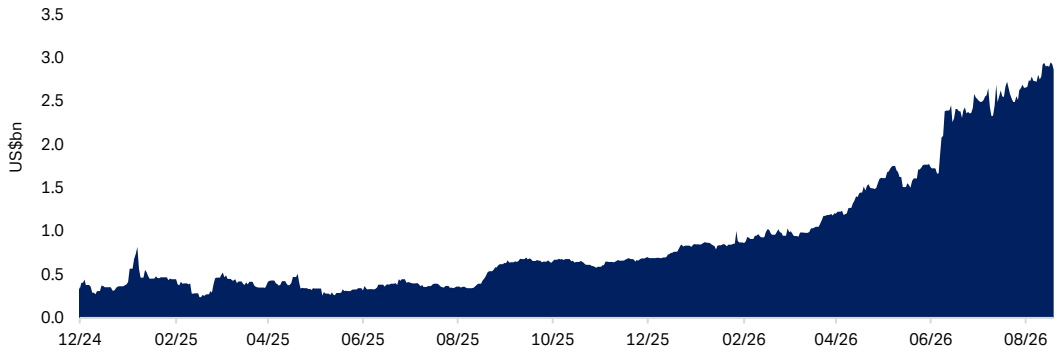
- DTCC's tokenization service, scheduled to launch in Oct, will allow eligible existing securities to move across blockchain infrastructure within its regulated framework.
- These tokenized securities will retain the entitlements, investor protections and ownership rights associated with their traditional form.
- NYSE has announced plans for a tokenized securities platform designed to support 24/7 trading of US-listed equities and ETFs, with near-instant settlement and stablecoin funding.

Benefits: Reduce barriers for foreign investors; automated post-trade processes and longer trading hours; reduce information asymmetry (e.g., embedding listing or disclosure rules directly into the tokens).

Risks & limitations

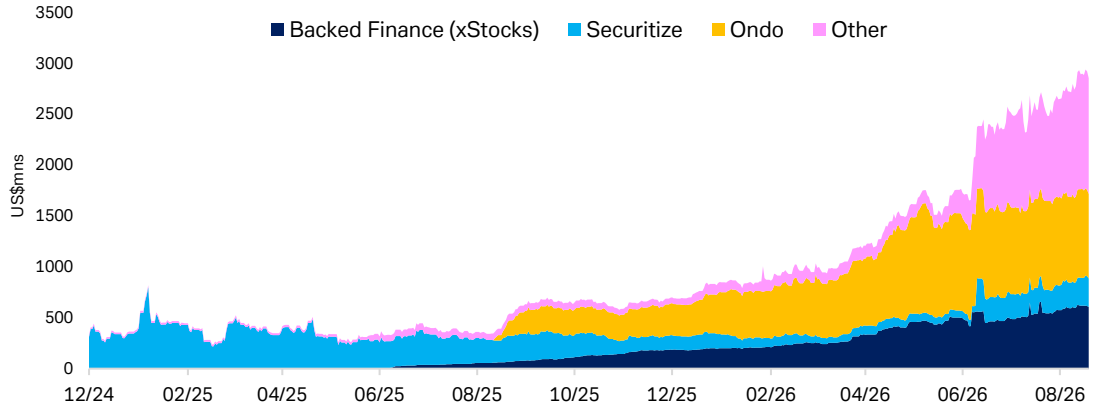
- Tokenized equities require clear legal ownership, enforceable investor rights and alignment with existing securities rules.
- Under some third-party wrapper structures, the token provides exposure to a share without making its holder a shareholder in the underlying company.
- Investors may therefore face issuer or custodian risk and may not receive the same voting, dividend or other rights as direct shareholders.

Market value of tokenized equities



Source: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

Tokenized equities by platform



Sources: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

LONG-TERM ADOPTION: Private credit expands through record-keeping; real estate is growing fast but from a negligeable base

Private credit

- Private credit stands ~$8bn in total, including asset-backed credit ($2.6bn), specialty finance ($2.5bn), corporate credit ($1.9bn), and diversified credit ($798mn).
- Activity includes lending linked to digital assets and selected corporate credit products.
- A larger use case is blockchain-based record-keeping: represented credit stands at ~$35bn, over 4x the distributed market.
- Companies like Figure Technologies and Tradable use blockchains to support loan administration, ownership records, payments tracking, and servicing.

Challenges

- Tokenization remains concentrated among a small number of issuers and lacks a deep, and liquid secondary market.
- Legal fragmentation, credit quality and servicing arrangements remain important risks. Blockchain records do not remove the underlying borrower's default risk.

Real estate

- The market expanded from ~$35mn in 2025 to $226mn in Sep 2026, although it remains small.

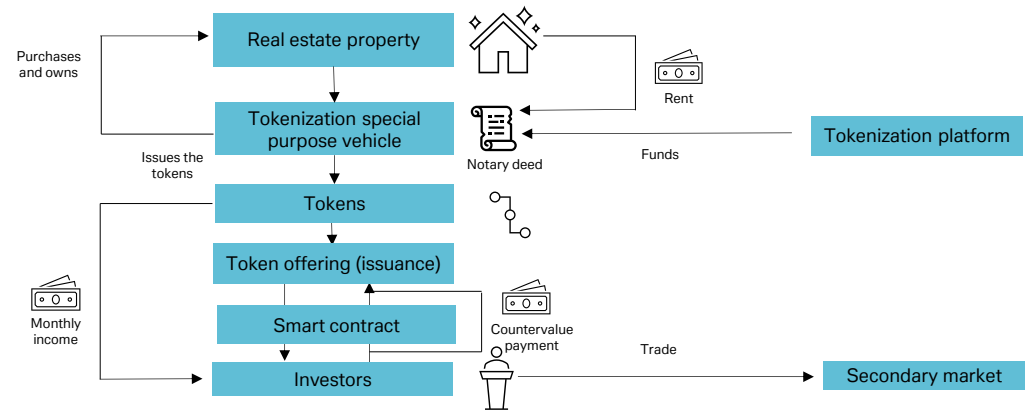
Benefits

- Fractional ownership can lower minimum investment sizes (typically for $100-$1,000 per unit) and widen access, particularly where traditional capital markets and mortgage markets are limited.

Challenges

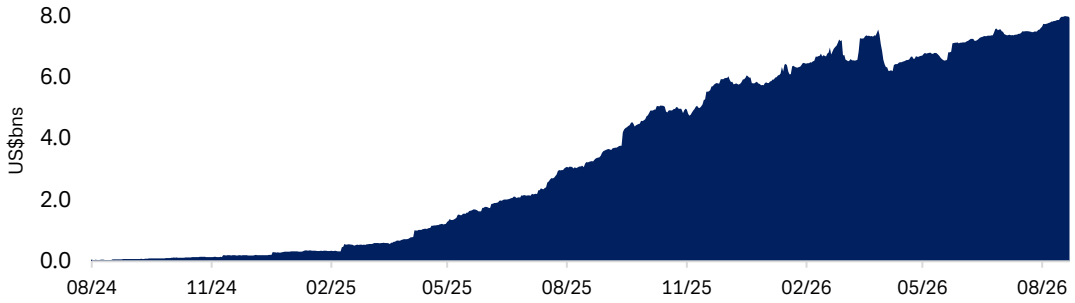
- A BIS study found that liquidity improvements can depend on platforms offering to buy tokens back. These arrangements can help investors exit but expose platforms to liquidity and solvency risks.

Real estate tokenization process



Source: Deutsche Bank Research, BIS, "When bricks meets bytes: does tokenization fill gaps in traditional real estate markets?"

Tokenized credit market stands at ~$8bn



Sources: Deutsche Bank Research, RWA.xyz. Updated September 10. Excludes platforms that issues tokens using blockchain as a recordkeeping layer (i.e. Represented Layer)

BARRIERS TO ADOPTION: Weak secondary market liquidity and lack of regulations continue to challenge adoption

Main barriers to adoption

- Secondary-market liquidity remains low. Issuance growth has not consistently translated into active trading or deep markets.
- Regulatory uncertainty: Fragmented or incomplete frameworks slow institutional adoption and complicate integration with traditional financial infrastructure.
  - Legal uncertainty persists over whether token ownership conveys ownership of the underlying asset or a claim on it.
  - In segments like money market funds, rapid cross-border growth can increase regulatory arbitrage, capital-flow volatility and AML/CFT compliance challenges. Policymakers must adapt frameworks as the technology evolves.
- Fragmented infrastructure: Public blockchains, permissioned institutional ledgers and hybrid platforms often lack interoperability, fragmenting markets. Clearstream, DTCC and Euroclear published a joint framework to address this in March 2026.
- Limited integration with existing custody, settlement, and record-keeping arrangements prevents institutions from scaling tokenized products.
  - Proprietary infrastructure and centralized controls also raise questions about ownership and governance.

Jurisdiction	Regulatory status	Main implication
	EU MiCA + DLT pilot regime	Comprehensive crypto-assets regulation. Tokenized financial instruments and tokenized deposits remain subject to securities and deposits regulation respectively (e.g. MiFID). National pilots (e.g., EU's DLT Pilot Regime) are testing tokenized capital-markets applications; focus is on capital market infrastructure modernization.
	UK Cryptoassets Regime / BoE Systemic Stablecoins Regime Live - Digital Securities Sandbox, Digital Gilt pilot Program (DIGIT) and Great British Tokenised Deposits (GBTD) Project	UK Cryptoassets and Stablecoins Regime finalised in 2026. FCA and BoE launched the Digital Securities Sandbox for firms to test DLT-based blockchains; HSBC first bank to go live. The FCA has also identified fund tokenization as a strategic priority through 2025-2030. FCA/BoE launched joint vision for wholesale tokenised markets in 2026 – work ongoing. DIGIT / GBTD development ongoing.
	GENIUS Act to begin in Jan 2027 with final rule out by Nov 26; CLARITY Act stalling	GENIUS establishes a federal framework for stablecoins. Key controversy is whether yield or reward mechanisms should be permitted in stablecoins and other digital assets; oversight is fragmented across the SEC, CFTC, banking regulators and state authorities. CLARITY Act, market structure bill, addressing yield proposal and related items failed its procedural vote in Sep. Attention will refocus on US federal regulators for digital asset rules in absence of market structure.
	China opposes permissionless crypto, not tokenization itself	Strict rules on cryptocurrencies remain; China's focus remains on e-CNY and state-approved blockchain networks. Recent reforms suggest the e-CNY is moving away from CBDC towards a tokenised deposits model, reflecting limited retail CBDC uptake alongside existing mobile payment platforms solutions.
	Project Ensemble Sandbox (Aug 2024); new regulatory roadmap for digital assets (Feb 2025); progress ongoing.	HKMA has supported tokenized green bonds; tokenized deposits; tokenized money-market and investment products. Project Ensemble has progressed to real-value transaction testing using tokenised deposits. HK to announce a new digital bond platform in H22026 to support issuance and settlement of tokenized bonds.
	Project Guardian (c. 2022)	Institutional tokenization sandbox; money market funds and bonds are tokenized and settled on-chain. Focus is increasingly on commercialisation and scaling successful pilot projects across funds, bonds, deposits and FX.
	Financial Instruments and Exchange Act, overseen by the Financial Services Agency	Legally recognizes security tokens through FIEA since 2020. Subjects security tokens (STOs) to the same regulations as traditional securities, including registration and disclosure, custody and investor protection obligations.
	Abu Dhabi's ADGM regulatory framework treats security tokens as regulated financial instruments.	Tokenized securities classified as "digital securities" under ADGM. Actively supports blockchain-based projects that allow businesses to test tokenization ideas via its sandbox program.

Sources: Deutsche Bank Research.

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OUTLOOK: Tokenized capital markets could become a core part of financial by the 2030s

- Tokenization will modernize financial infrastructure rather than eliminate intermediaries. We expect a gradual transition onto digital rails, with banks retaining key roles across payments, FX, treasury, financing, collateral and trading settlement.

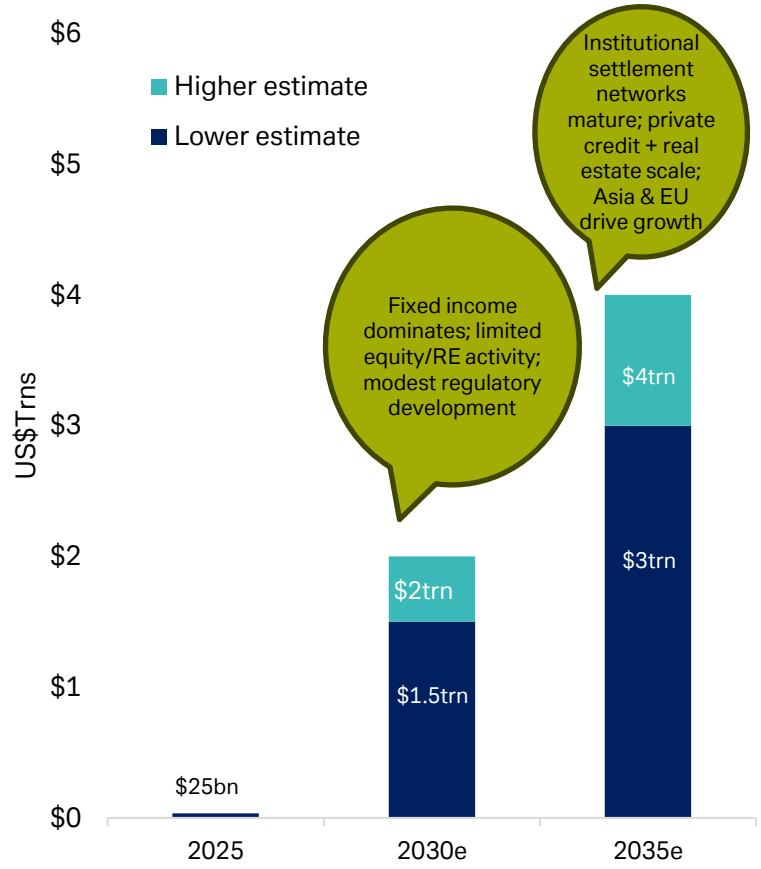
Assumption, on the positive side reflects…

- So far, tokenization has achieved its greatest scale in stablecoins, with activity concentrated in crypto trading and payments gaining ground.
- Short-term: Adoption will be driven by tokenized US Treasuries and money market funds, expanding as yield-bearing collateral in repo and margin arrangements.
- Long-term: The opportunity extends beyond assets to the infrastructure supporting issuance, settlement, custody, collateral management and repo markets.
- We are seeing 3 growth catalysts: (i) infrastructure: major financial market infrastructures (DTCC, NYSE, Nasdaq) are integrating tokenization into core workflows; (ii) liquidity: Growth of regulated on-chain money (stablecoins and tokenized deposits) providing a trusted settlement foundation; (iii) and regulation: US 2025 GENIUS ACT continues to encourage global digital asset adoption, although the failure of the CLARITY act to advance yesterday likely rules out progress in this congressional session until Jan 2027. Looking ahead, focus will turn back to the SEC and CFTC for crypto rules.

…Risks to adoption

- Accelerated liquidity runs: Tokenization may create a new form of systemic liquidity risk that spreads faster than traditional finance.
- Money market funds yield-chasing: Declining interest rates or rising competition may push MMF issuers into risker assts, less-liquid assets to boost yield, undermining their safety.
- Crypto contagion: Tight integration with crypto markets via stablecoin reserves and collateral risks transmitting digital asset volatility to conventional financial markets.
- Operational risks: Smart-contract vulnerabilities, cyberattacks, network outages could halt trading and prevent access to the underlying assets.
- Financial Crime: Traditional institutions must scale up compliance and AML controls to manage illicit activities on decentralized networks.

Conservative forecasts for tokenized real-world assets (ex stablecoins)



Source: Deutsche Bank Research.

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Appendix 1



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